AYMANE AIT DADS

ML Engineering Intern - LLM Research & Development | LLM Fine-Tuning · PyTorch · Open-Source Models

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PROFESSIONAL SUMMARY

Fine-tuned a 30B Mamba-Transformer MoE model (Nemotron) with LoRA on a $106K+ Kaggle competition, reaching top 10% public leaderboard (0.80/1.0) through 20+ single-variable ablation studies and production-grade GPU engineering on Blackwell hardware. Ranked 1st in EURECOM class on an AI-generated image detection benchmark (0.791 AUC) using CLIP ViT-L/14 fine-tuning, and submitted to NTIRE 2026 @ CVPR CodaBench international benchmark. Brings hands-on experience with PyTorch, Hugging Face Transformers, PEFT, TRL, vLLM, and the full ML lifecycle from training runs to model serving and evaluation. Available for a 6-12 month internship in Amsterdam with no work authorization barrier (EU-based, France).

CORE COMPETENCIES

LLM Fine-Tuning | Open-Source LLMs | PyTorch & Hugging Face | Experiment Design & Evaluation | Production Serving Infrastructure | Full ML Lifecycle | Distributed Training | A/B Testing & Ablation Studies

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Trained Random Forest and K-Means models on **100K+ fiber-optic network records** to classify network quality across thousands of nodes, directly supporting the network optimization team.
- Designed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, delivering output adopted immediately by production network planning workflows.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines built in Python and Scikit-learn.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, **visualized in Power BI** dashboards for non-technical executives.

PROJECTS

NVIDIA Nemotron Reasoning Challenge *Kaggle Competition · In Progress (Jun 2026) · Top 10% Public Leaderboard · $106K+ Prize Pool*

- Fine-tuned a **30B Mamba-Transformer MoE open-source LLM** (3.5B active params via MoE routing) with LoRA (r=32, ~888M trainable parameters), reaching 0.80/1.0 on the public leaderboard (top 10% of $106K+ competition) via SFTTrainer completion-only loss on 7,828 chain-of-thought reasoning examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patching caching allocator warm-up to prevent 120GB OOM, stubbing mamba3/cutlass imports, and replacing rmsnorm_fn with a pure PyTorch fallback - then ran 20+ single-variable ablations confirming packing and synthetic data consistently hurt generalization.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in EURECOM Class*

- Fine-tuned **CLIP ViT-L/14 (428M params)** with a custom MLP head and dual learning rates, achieving 0.791 AUC on the private Kaggle leaderboard to rank 1st in the EURECOM class, with a separate submission to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Designed a **3-model ensemble with 10-view TTA** (flips, crops, blur, resizes) trained on 250K images across 25+ generator types using FP16 mixed precision, and identified via ablation that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) due to superior out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya *2021 - 2023*

SKILLS

LLM & ML Engineering: PyTorch | Hugging Face Transformers | LoRA / QLoRA | PEFT | TRL / SFTTrainer | vLLM | Unsloth

ML & Computer Vision: Scikit-learn | OpenCLIP | CLIP ViT | CNN / DenseNet | TensorFlow | Pandas | NumPy

MLOps, Infra & Languages: Python (expert) | Git / Linux | FP16 mixed precision | CUDA / Triton debugging | Docker | SQL | Bash